

Supplementary Materials for Ego-Human Motion Prediction with 3D-Aware LLM

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A Overview

Our supplementary material provides further clarification of the proposed method and offers additional insights and in-depth discussions on areas not extensively covered in the main paper:

- Implementation Details (Sec. B).
- Additional Experimental Results (Sec. C).
- Additional Qualitative Results (Sec. D).

B Implementation Details

B1 Model Architecture and Training Details

We use GPT-2 medium and Qwen-2.5 0.5B models for language model backbones. For motion tokenization, we use a product-quantized VAE (PQ-VAE) following EgoLM, employing a convolutional encoder-decoder architecture with a temporal downsampling rate of 4, a codebook of 4096 entries with 64-dimensional codes, and exponential moving average updates with codebook reset [4]. We use two codebooks for the PQ-VAE. Motion sequences are sampled at 10Hz and have lengths between 45 and 300 frames. For scenes, we retain maximum 10,000 points after voxelization and sampling. The Q-Former is initialized from pretrained BERT-base [3] and uses 32 learned query tokens. For modality-specific projection layers, we use a linear layer for 3D scene embeddings, a small MLP for three-point tracking features, and another small MLP for video frame embeddings. The three-point tracking features are constructed by concatenating the 3D translation, 6D rotation, 3D translation velocity, and 6D rotation velocity for all three points along the channel dimension, resulting in a 54-dimensional representation.

In both pre-training and instruction tuning, we train with AdamW [9] optimizer (learning rate 1×10^{-4}) and a cosine-annealing learning rate scheduler [8], using a batch size of 16. During the multi-modal reward GRPO-based reinforcement finetuning, G is set to 6, and the reward weights are established at $w_m = 1.0$, $w_t = 0.8$, and $w_d = 0.02$. The clipping parameter ϵ and KL coefficient β are assigned values of 0.2 and 0.001, respectively. Regarding R_{format} , a score of 0

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is granted if the generated output follows the structure described in Sec. B4; otherwise, a penalty of up to -3 is applied. Finally, the learning rate and batch size are set to 1×10^{-6} and 4, respectively. Our model runs at 33.9 ms time-to-first-token and 302.1 ms total inference latency for future motion prediction on a single NVIDIA A6000 GPU.

B2 Pose Representation Details

Human motion is modeled as a kinematic tree of body joints with a root joint that carries global translation and rotation over time. We adopt a kinematic tree comprising 23 joints and a 6D rotation representation for stable regression while capturing dynamics. At time t , the pose state comprises (i) root translational velocity $V_t^r \in \mathbb{R}^3$, (ii) root rotation $R_t^r \in \mathbb{R}^6$, (iii) root rotational velocity $R_t^{rv} \in \mathbb{R}^6$, (iv) joint rotations $R_t^j \in \mathbb{R}^{22 \times 6}$, and (v) joint rotational velocities $R_t^{jv} \in \mathbb{R}^{22 \times 6}$. Concatenating these terms yields a vector $M_t = [V_t^r, R_t^r, R_t^{rv}, R_t^j, R_t^{jv}] \in \mathbb{R}^{279}$, and a sequence $M = \{M_t\}_{t=1}^T$ represents a motion clip. Joint positions $J \in \mathbb{R}^{23 \times 3}$ are recovered via forward kinematics from $\{M_t\}$.

B3 Pretraining Templates

For spatial-semantic scene awareness LM pretraining, we design a pair of prompt templates for spatial and semantic reasoning to enhance scene understanding. The exact templates are as follows. The 3D scene and video embeddings shared by all tasks are injected via $\langle 3D_Scene_Placeholder \rangle$ and $\langle Video_Placeholder \rangle$. The natural language query and response are placed at $\langle Question_Placeholder \rangle$ and $\langle Answer_Placeholder \rangle$, respectively. These placeholders standardize how *scene and video embeddings* and *text* are concatenated and fed into the LM, enabling consistent training across pretraining objectives.

Spatial Scene Awareness

Input: *Answer the question with $\langle 3D_Scene_Placeholder \rangle$ scene and the corresponding $\langle Video_Placeholder \rangle$ video frames: What obstacles are around the person in the current frame?*

Output: $\langle Answer_Placeholder \rangle$

Semantic Scene Awareness

Input: *Answer the question with $\langle 3D_Scene_Placeholder \rangle$ scene and the corresponding $\langle Video_Placeholder \rangle$ video frames: $\langle Question_Placeholder \rangle$*

Output: $\langle Answer_Placeholder \rangle$

B4 Instruction Tuning Templates

The instruction template for a unified instruction tuning objective that jointly outputs motion tracking, motion forecasting, past motion description, and future

motion description is shown below. The 3D scene embeddings and video embeddings are inserted into $\langle 3D_Scene_Placeholder \rangle$ and $\langle Video_Placeholder \rangle$, respectively.

Similarly, the embeddings for three-point motion tracking are placed at $\langle TP_Placeholder \rangle$. The output targets, including the spatial reasoning answer, past motion, future motion, past motion description, and future motion description, are tokenized and inserted into their corresponding placeholders. Delimiter tokens are used to structure prompts across tasks and modalities, but are omitted from the template below for simplicity.

Multi-Modal Multi-Task Instruction Tuning

Instruction: *Perform human motion tracking and prediction. First, analyze the spatial constraints (front, left, right) and the optimal direction from the scene. Then, output current and future human motion token sequences, and describe current and future human motion conditioned on the given scene, observed video CLIP embeddings, and observed three-points features.*

Input scene: $\langle 3D_Scene_Placeholder \rangle$.

Input video CLIP embeddings: $\langle Video_Placeholder \rangle$.

Input three-points features: $\langle TP_Placeholder \rangle$.

Output: $\langle Answer_Placeholder \rangle \langle Past_Motion_Placeholder \rangle \langle Future_Motion_Placeholder \rangle \langle Past_Description_Placeholder \rangle \langle Future_Description_Placeholder \rangle$

B5 3D Scene QA Dataset Generation

Semantic Awareness ($\mathcal{Q}_{sem}$) Dataset Generation. We construct a semantic awareness QA dataset using Qwen2.5-VL-7B [2]. Given a 3D scene, we sample a short sequence of egocentric frames from each video segment that observes the scene and feed these frames to the model with an instruction prompt as shown below. For every frame, the model produces k QA items, which we later aggregate over all segments to form a single QA set per scene. The instruction includes (i) a target distribution over QA types (*e.g.* object, color, object nature, place, number, other), (ii) specificity rules that reduce ambiguity by requiring concrete object names, informative modifiers (color/material/shape/size/state), and explicit locator phrases (relation + anchor/reference), and (iii) constraints that exclude humans and human-related attributes. For clarity and consistency, the prompt also provides a few illustrative examples for each QA type. Finally, all QA items generated for the segments associated with a scene are merged and lightly filtered to yield the scene-level semantic QA set. Representative examples of the resulting QA sets are shown in Fig. A.

Semantic Awareness Dataset Generation

Instruction: *Given the following frames of a 3D scene, generate question-answer (QA) items about the scene. Create k items in total.*

Requirements:

- QA type distribution goal: 30% object, 10% color, 10% object nature, 30% place, 10% number, 10% other.
- QA specificity: Use concrete object names and, when helpful, modifiers (color/material/shape/size/state) and explicit locator phrases (relation + anchor/reference) to uniquely identify the target.
- ⋮
- Exclude humans: If humans appear in the frame, ignore them and focus on non-human objects.

Examples:

- **Object**
 Q: "What is on <Anchor> beside <Reference>?"
 A: "<Target> "
 Q: "What is on <Anchor> to the left of <Reference>?"
 A: "<Target> "
- **Place**
 Q: "Where is <Target>?"
 A: "On <Anchor> beside <Reference> "
 Q: "Where is <Target> on <Anchor>?"
 A: "Left of <Reference> "
- **Color**
 Q: "What color is <Target>?"
 A: "<Color> "
- **Object nature**
 Q: "What material is <Target>?"
 A: "<Material> "
 Q: "What state is <Target>?"
 A: "<State> "
- **Number**
 Q: "How many <Target> are near <Reference>?"
 A: "<Number> "
- **Other**
 Q: "Is there a <Target> on <Anchor>?"
 A: "<Yes / No> "

Output: k QA items

Spatial Awareness ($\mathcal{Q}_{\text{spa}}$) Dataset Generation. The spatial awareness dataset characterizes the navigatable environment around the human by quantifying directional clearance and collision risk. For each observation frame, the dataset provides categorical free-space labels for the front, left, and right directions, corresponding binary collision flags, and a best direction label indicating the least obstructed direction available for locomotion.

Each scene is represented by a set of voxels, a grid origin coordinate, and a uniform voxel resolution of 0.15m. The head pose from three point tracking provides, for every temporal frame, a 3D translation vector and a 3×3 rotation matrix. Dataset generation proceeds in two stages. First, a proximity filter

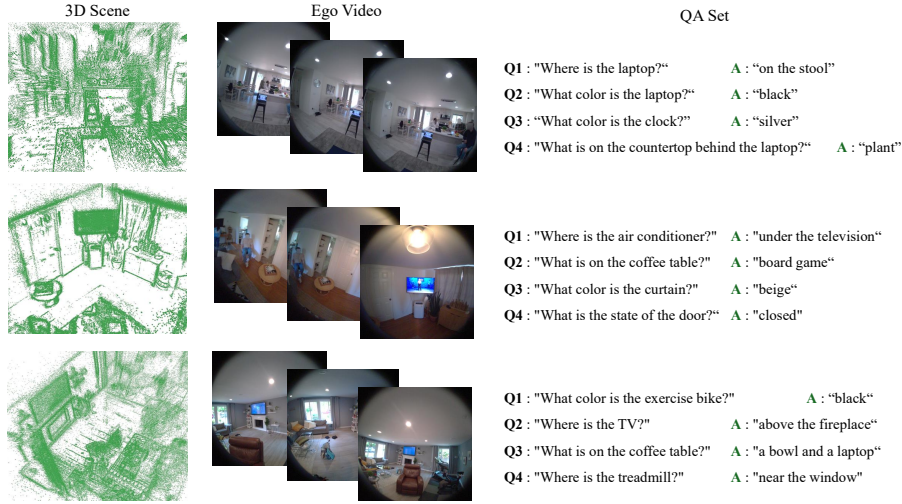
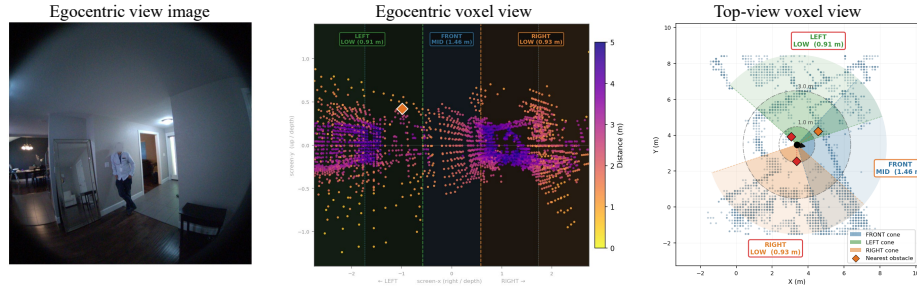


Fig. A: Examples of generated semantic awareness dataset. 3D scene (left), corresponding egocentric video frames (middle), and example QA items (right). The resulting dataset comprises rich contextual samples that are capable of training the LM to learn the 3D semantics.

retains only scene voxels whose vertical offset from the head position falls within $\pm 1.0\text{m}$, restricting the search to body relevant geometry. Second, for each of the four directions (front, left, right, and back), a cone query selects voxels whose horizontal bearing lies within a cone with a half angle of 60° , and the minimum Euclidean distance to the selected voxels defines the free space distance for that direction. Free-space distances are discretized into three categorical levels: **LOW** ($d < 1.0\text{m}$), **MID** ($1.0 \leq d < 3.0\text{m}$), and **HIGH** ($d \geq 3.0\text{m}$). A complementary collision check simulates a 0.5m step in each direction and tests whether any voxel falls within a cylindrical body proxy of radius 0.3m and half-height 1.0m centered at the projected position. The direction with the greatest free space distance is designated as the best direction label (**BEST_DIR**). The resulting caption summarizes this information in a single sentence, as shown in Fig. B. Full algorithmic details are provided in Algorithm 1.

B6 Data Split

We conduct training and evaluation on the Nymeria dataset [10], using the provided human motion (3D pose), three-point trackings, motion descriptions, 3D scene point clouds, and egocentric video frames. As Nymeria *does not offer an official split*, we construct splits based on scenes containing narration annotations. We stratify scenes by dominant motion pattern (largely static vs. dynamic human motion) to maintain a similar static/dynamic ratio in both splits, based on Nymeria’s scene activity tags (action scripts); for example, static activities include



Caption: “**Front** is clear with **MID** clearance. **Left** is blocked with **LOW** clearance. **Right** is blocked with **LOW** clearance. The optimal direction to move is **front**.”

Fig. B: Visualization of spatial awareness dataset generation. For each egocentric frame, the surrounding environment is parsed into three directional cone sectors (front, left, right) over the height-filtered 3D voxel reconstruction, and the minimum free-space distance within each cone is discretized into categorical clearance levels (**LOW**, **MID**, **HIGH**) based on the distance to its closest obstacle in each cone sector. The resulting spatial caption describes the clearance state and optimal navigation direction, providing structured supervision for the LM to internalize obstacle layouts.

simon says, *laundry*, and *cooking*, while dynamic activities include *housekeeping*, *welcome to my place*, and *where is X*. We set a ratio of 4:1 between training and validation scenes. The exact split specification will be released alongside our code.

B7 Baseline Models

EgoLM. We adapt EgoLM [6], originally designed for egocentric motion tracking and narration, to motion forecasting. Because EgoLM does not directly support forecasting instructions, we instead use its motion-to-motion (M2M) pretraining, in which the language model backbone is trained to autoregressively generate motion sequences.

We consider two EgoLM baselines that differ in how the past motion condition is obtained. **EgoLM (GT motion)** uses the M2M pretrained checkpoint and conditions forecasting on the ground truth past motion. **EgoLM (Inst. tuning)** first uses the instruction tuned EgoLM checkpoint to reconstruct past motion from sparse three point tracking, and then feeds the reconstructed past motion to the same M2M pretrained checkpoint to forecast the future.

For both variants of the inference, directly prompting the model with the full past sequence often leads to early termination and truncated outputs. To mitigate this implicit length bias and generate the full prediction horizon, we use a recursive two-step generation strategy. We first condition on the latter half of the past motion to generate the first half of the future motion, and then condition on the generated segment to produce the remaining second half of the future.

Ffiction. Ffiction [1] originally predicts future motion from past motion, given past sequence of egocentric video and 3D scene represented as voxel. We modify

Algorithm 1 Spatial Awareness Dataset Generation

Require: Scene point cloud $\mathcal{S} \subset \mathbb{R}^3$, head position $\mathbf{p} \in \mathbb{R}^3$, rotation matrix $\mathbf{R} \in \text{SO}(3)$, and horizontal projection Π

Ensure: Labels $\{\text{FREE_FRONT}, \text{FREE_LEFT}, \text{FREE_RIGHT}, \text{COLLIDE_FRONT}, \text{COLLIDE_LEFT}, \text{COLLIDE_RIGHT}, \text{BEST_DIR}\}$

- 1: $\mathbf{f} \leftarrow \Pi(\mathbf{R}[:, 1])$, $\mathbf{r} \leftarrow \Pi(\mathbf{R}[:, 0])$ $\triangleright$ Horizontal projections of forward/right axes
- 2: $\mathcal{D} \leftarrow \{\mathbf{f}, -\mathbf{r}, \mathbf{r}, -\mathbf{f}\}$ $\triangleright$ FRONT, LEFT, RIGHT, BACK unit vectors
- 3: $\mathcal{S}' \leftarrow \{\mathbf{s} \in \mathcal{S} : |s_z - p_z| < \delta_h\}$ $\triangleright$ Height filter: $\delta_h = 1.0$ m
- 4: **for** each direction $\mathbf{d} \in \mathcal{D}$ **do**
- 5: $\mathbf{r}_i \leftarrow \mathbf{s}_i - \mathbf{p}$, $d_i \leftarrow \|\mathbf{r}_i\| \quad \forall \mathbf{s}_i \in \mathcal{S}'$
- 6: $\hat{\mathbf{r}}_i^h \leftarrow \Pi(\mathbf{r}_i) / \|\Pi(\mathbf{r}_i)\|$ $\triangleright$ Horizontal unit direction to voxel
- 7: $\mathcal{C} \leftarrow \{i : \hat{\mathbf{r}}_i^h \cdot \mathbf{d} > \cos \theta_c, 0.05 < d_i < d_{\max}\}$ $\triangleright$ Cone filter: $\theta_c = 60^\circ$
- 8: $\text{free}(\mathbf{d}) \leftarrow \min_{i \in \mathcal{C}} d_i$ (or $d_{\max}$ if $\mathcal{C} = \emptyset$)
- 9: $\text{category}(\mathbf{d}) \leftarrow \begin{cases} \text{LOW} & \text{if } \text{free}(\mathbf{d}) < 1.0 \\ \text{MID} & \text{if } 1.0 \leq \text{free}(\mathbf{d}) < 3.0 \\ \text{HIGH} & \text{otherwise} \end{cases}$
- 10: $\mathbf{p}' \leftarrow \mathbf{p} + \Delta s \mathbf{d}$ $\triangleright$ Simulated step: $\Delta s = 0.5$ m
- 11: $\text{collide}(\mathbf{d}) \leftarrow \exists \mathbf{s} \in \mathcal{S} : \|\Pi(\mathbf{s} - \mathbf{p}')\| < r_b \wedge |s_z - p'_z| < \delta_h$ $\triangleright r_b = 0.3$ m
- 12: **end for**
- 13: $\text{BEST_DIR} \leftarrow \arg \max_{\mathbf{d} \in \mathcal{D}} \text{free}(\mathbf{d})$
- 14: **return** categorical labels and collision flags for FRONT/LEFT/RIGHT, BEST_DIR

this framework to our setting, namely: (i) predicting future motion from past three-point tracking data and (ii) tracking past motion from three-point tracking data, where egocentric video and 3D scene representations are provided. Therefore, we modify the encoder to encode three-point tracking data instead of the past motion via an MLP network. Also, we additionally add a past motion tracking head decoder that still samples based on CVAE, as in the original future pose prediction head.

UniEgoMotion. UniEgoMotion [13] is a unified conditional motion diffusion model that reconstructs, forecasts, and generates scene-aware human motion from egocentric images alone. We re-implement UniEgoMotion to take 3D scene and three-point tracking data as input in a similar manner as FIction and our Ego3DLM. Also, past egocentric video is given as input instead of single image.

LLM Narration. As no works exist that predict future motion in text modality as in our predictive setting, we compare with the zero-shot capability of Qwen2.5-VL-7B [2]. We provide uniformly sampled 10 egocentric video frames from the past sequence and prompt the model to predict the future motion description with the following prompt.

Instruction: You are given RGB frames from sequence $\langle \text{source_seq} \rangle$ of scene $\langle \text{scene_name} \rangle$.
 Predict what is most likely to happen in the immediate next sequence $\langle \text{target_seq} \rangle$.

Focus on motion and scene dynamics, not static descriptions.

Study the following ground-truth style examples and mimic their tone, tense, and specificity: <examples_block>

Style constraints:

- *Always describe the actor as “C” (e.g. “C is standing in the hallway...”).*
- *Use present/progressive tense to mirror GT descriptions (avoid “will/going to”).*
- *Include concrete verbs and referenced objects/surfaces whenever visible.*
- *1-3 sentences maximum; no bullet lists.*

For the <examples_block>, we provide the following GT sequences to provide context for the LLM to generate responses with a similar style of text as the GT:

- *“C is standing in the hallway, then leans to her right to reach for the door before opening it with her right hand.”*
- *“C walks into the kitchen, rests her left hand on the counter, and bends to peer into the refrigerator.”*
- *“C steps toward the sofa, lowers herself into a seated position, and keeps talking to the person next to her.”*

B8 Evaluation Metric

All metrics calculated in the embedding space (FID, Diversity, d_{gp} , d_{pg} , d_{pp}) are first obtained the motion and text embedding with respective encoders trained as in [5] on our training split of the Nymeria dataset.

Common Motion Metric. As local motion plausibility plays a critical role in both motion forecasting and tracking, we employ APE as a shared metric to effectively measure the local positional accuracy of the generated motions.

- **APE (Aligned mean per-joint Position Error):** APE measures local pose accuracy as the mean ℓ_2 distance between predicted and ground-truth joint positions after aligning the head position of both predicted and ground-truth skeletons, evaluating articulated motion independent of global translation.

Motion Forecasting Metric. For the future motion prediction, we generate three modes and select the best one based on the lowest JPE, since it comprehensively captures both global translation and local joint articulation errors. To evaluate joint positional accuracy, we calculate APE, JPE, and ADE, alongside FDE to assess the precision of the final displacement. For both ADE and FDE, we also provide short-term evaluations restricted to the first 2 seconds. Additionally, we compute FID and Diversity to evaluate the statistical realism and variation of the generated motion distributions.

- **JPE (Joint Position Error):** JPE measures global pose accuracy as the mean ℓ_2 distance between predicted and ground-truth joint positions in the world frame, penalizing errors in both body placement and joint configuration.

- **ADE (Average Displacement Error of head position):** ADE is the average Euclidean distance between predicted and ground-truth head positions over all forecast timesteps, analogous to the standard ADE metric in trajectory prediction. ADE_{2s} refers to ADE of the first 2 seconds.
- **FDE (Final Displacement Error of head position):** FDE is the Euclidean distance between predicted and ground-truth head positions at the final forecast timestep, emphasizing long-horizon accuracy. FDE_{2s} refers to FDE at 2 seconds.
- **FID (Fréchet Inception Distance on motions):** FID is computed between multivariate Gaussian fits of ground-truth and generated motion embeddings (random 300 validation samples), with lower values indicating a closer match between real and generated motion distributions.
- **Diversity (motion embedding diversity):** Diversity is the mean ℓ_2 distance between randomly sampled pairs of motion embeddings (up to 300 pairs), where higher values indicate a richer, more varied set of generated motions.

Motion Tracking Metric. For motion tracking evaluation, we evaluate positional accuracy using the overall APE, as well as the regional Upper and Lower body APE. Furthermore, we assess rotational accuracy by calculating the root joint angular error and the mean rotation error across all articulated joints.

- **Upper and Lower (Upper & Lower body APE):** Upper and Lower measure local pose accuracy by calculating the mean APE for specific body parts. Upper is the mean APE of 14 joints, including the root joint and those located above it, while Lower is the mean APE of 10 joints, including the root joint and those located below it.
- **Root (Root Angle):** Root angle is the mean geodesic angular error between the root orientations of the head-aligned predicted past motion and the ground-truth past motion, reflecting how accurately the pelvis/torso facing direction is recovered.
- **J.A. (Joint Angle):** Joint angle is the mean rotation error over all articulated joints: each child-parent bone is converted to a local 6D rotation and compared with the corresponding ground-truth joint frame, measuring how well limb and joint orientations match independently of the root pose.

Text Metric. In motion description evaluation, we use widely adopted metrics for generated text quality, including Bleu, Rouge_L, SBert, and R@N.

- **Bleu and Rouge_L:** Bleu [12] measures n -gram overlap between predicted and ground-truth texts to quantify lexical precision, scaled from 0 to 1. Rouge_L [7] evaluates the Longest Common Subsequence (LCS) between predicted texts and the ground truth, effectively capturing similarities in word order and overall sentence structure. Please refer to the corresponding papers for detailed mathematical formulations.

- **SBert (SBert similarity between sentences)**: SBert [14] measures the semantic alignment between predicted and ground-truth texts by computing the mean cosine similarity of their sentence embeddings encoded via a pre-trained SBERT model.
- **R@N (top-N R-precision accuracy)**: R@N evaluates the Euclidean distances between predicted text embeddings and ground-truth motion embeddings within a fixed-size subset of size 300. It represents the proportion of instances where the true corresponding ground-truth motion embedding is ranked within the top N closest matches to the predicted text embedding, with higher values indicating stronger semantic alignment between the text and motion modalities.

Motion-description Alignment Metric. To evaluate the semantic consistency and cross-modal alignment between textual descriptions and generated motions, we utilize narration-motion matching distances.

- d_{gp} , d_{pg} , d_{pp} (**Narration-motion matching distance**): Narration-motion matching distance measures the cross-modal alignment as the mean ℓ_2 distance between narration and motion embeddings. It evaluates the semantic correspondence across different pairings: ground-truth narration versus predicted motion (d_{gp}), predicted narration versus ground-truth motion (d_{pg}), and the self-consistency of jointly predicted narration-motion pairs (d_{pp}).

C Additional Experimental Results

In this section, we provide additional analyses of our design choices and the effects of 3D scene conditioning. We compare baselines with and without 3D scene inputs, and ablate joint decoding across motion and text as well as across past and future tasks, highlighting consistent gains from unified generation. To further validate our approach, we strengthen the EgoLM baseline with dedicated motion forecasting instruction tuning, and verify that our gains persist on a modern LLM backbone in addition to GPT-2. We further assess the physical plausibility of our predictions through foot sliding and foot contact metrics, and confirm the faithfulness of our EgoLM reproduction by matching the original paper’s evaluation protocol. We also evaluate long-term forecasting and multi-hypothesis future prediction to assess effectiveness across extended horizons and generation diversity. Lastly, we evaluate the advantages of using GRPO and the generalization result on a different dataset. Ego3DLM[†] denotes our model trained up to instruction tuning only, without the GRPO reinforcement learning stage, to directly isolate and clarify the impact of our core model design.

Additional Baselines with/without 3D Scene Conditioning. Table A evaluates all methods with and without 3D scene input on motion prediction and tracking. While all baselines show modest JPE improvements from 3D scene conditioning, with Fiction reducing JPE by 3.9% and UniEgoMotion by 3.5%, these gains are limited by the absence of dedicated scene understanding, as scene

Table A: Ablation on the effect of 3D scene conditioning across all methods on motion prediction and tracking. While all methods benefit from 3D scene input, Ego3DLM[†] achieves the largest and most consistent gains.

Method	3D Scene	Motion Prediction (3 modes)						Motion Tracking				
		APE ↓	JPE ↓	ADE ↓	FDE ↓	FID ↓	Diversity ↑	APE ↓	Upper ↓	Lower ↓	J.A. ↓	Root ↓
Fiction		209.5	587.5	582.5	949.6	0.3691	0.7381	187.5	117.6	285.7	32.94	35.11
	✓	206.2	564.7	558.3	904.3	0.3275	0.7432	181.1	114.0	282.3	31.28	34.61
EgoLM (Inst. tune)		184.9	579.4	552.6	963.6	0.2137	0.6142	161.9	95.6	265.6	33.63	24.01
	✓	174.2	493.1	471.2	925.2	0.2184	0.5176	148.5	83.7	232.6	34.14	30.64
UniEgoMotion		157.5	439.5	430.4	740.6	0.1494	0.9171	156.2	82.7	248.5	28.52	23.96
	✓	151.5	424.3	409.7	720.5	0.1530	0.9022	152.2	79.4	233.3	26.48	22.71
Ego3DLM [†]		156.2	388.5	363.9	678.4	0.0426	0.9352	105.8	57.5	168.8	24.03	20.92
	✓	148.5	368.3	346.3	645.4	0.0190	0.9358	98.3	53.4	155.3	22.52	19.62

Table B: EgoLM trained with motion forecasting instruction tuning, under forecasting-only and tracking-and-forecasting multi-task settings. Although direct forecasting training strengthens the EgoLM baseline, Ego3DLM[†] still outperforms it across all metrics by effectively leveraging explicit 3D scenes.

Method	Motion Prediction (3 modes)			Motion Tracking	Future Motion Description	Past Motion Narration	x - y align.
	JPE ↓	ADE _{2s} ↓	FDE _{2s} ↓	APE ↓	Bleu-4 ↑	Bleu-4 ↑	d_{pp}^f ↓
EgoLM (fore.)	416.3	244.5	369.4	-	0.0766	-	6.94
EgoLM (track. & fore.)	423.5	253.7	379.3	194.8	0.0702	0.0891	7.66
Ego3DLM [†]	368.3	208.8	317.9	98.3	0.1036	0.1126	5.91

features are injected without prior spatial or semantic grounding. In contrast, Ego3DLM[†] achieves a larger JPE reduction of 5.2% with scene conditioning and reaches a JPE of 368.3, a 13.2% improvement over the strongest baseline, UniEgoMotion with 3D scene (424.3). We attribute this gain to our spatial-semantic pretraining, which explicitly trains the model to reason about object-level geometry and semantics before motion prediction is introduced, enabling the model to translate 3D scene geometry into physically informed and spatially coherent body motion predictions.

EgoLM with motion forecasting instruction tuning. In Table B, we directly train EgoLM for the motion forecasting task and make two variants: 1) Forecasting only, and 2) Multi-task with tracking and forecasting. Direct forecasting training indeed yields a stronger EgoLM baseline; nonetheless, our Ego3DLM remains superior across all metrics through its effective use of explicit 3D scenes.

Modern LLM backbone. While we have used GPT-2 for a fair comparison with EgoLM, we have implemented our method on a modern backbone Qwen 2.5-0.5B (Table C). We observed that Ego3DLM (Qwen 2.5) shows comparable motion tracking and prediction performance in Stage II. Notably, with GRPO, it surpasses Ego3DLM (GPT-2) on motion metrics, driven by a compelling GRPO gain.

Table C: Effect of a modern LLM backbone and GRPO fine-tuning. Beyond GPT-2, we implement Ego3DLM on a modern backbone (Qwen 2.5-0.5B). Applying GRPO yields consistent gains across metrics for both backbones, with the Qwen variant surpassing its GPT-2 counterpart on motion metrics.

Method	Prediction (3 modes)			Tracking	Future Motion Description	Motion Narration	x - y Align.	
	JPE ↓	ADE _{2s} ↓	FDE _{2s} ↓	APE ↓	Bleu-4 ↑	Bleu-4 ↑	d_{pp}^f ↓	d_{pp}^p ↓
GPT-2-medium	368.3	208.8	317.9	98.3	0.1036	0.1126	5.91	4.28
GPT-2-medium + GRPO	364.5 (+3.8)	205.9 (+2.9)	312.6 (+5.3)	96.4 (+1.9)	0.1039 (+0.0003)	0.1107 (-0.0019)	5.80 (+0.11)	4.25 (+0.02)
Qwen 2.5-0.5B	372.6	209.6	320.0	96.3	0.0915	0.1043	6.63	4.85
Qwen 2.5-0.5B + GRPO	361.7 (+10.9)	202.5 (+7.0)	309.6 (+10.3)	95.9 (+0.5)	0.0938 (+0.0023)	0.1057 (+0.0014)	6.36 (+0.27)	4.72 (+0.12)

Table D: Ablation study on joint vs. separate motion and text decoding. Joint decoding consistently improves motion prediction accuracy and future motion description quality, while also strengthening motion-description alignment.

Method	Motion Prediction (3 modes)						Future Motion Description				x - y Align.	
	APE ↓	JPE ↓	ADE ↓	FDE ↓	FID ↓	Diversity ↑	Bleu-4 ↑	Rogue _L ↑	R@3 ↑	R@1 ↑	d_{pg} ↓	d_{pp} ↓
Ego3DLM [†] (sep. motion/text)	149.5	370.2	348.7	660.3	0.0116	1.0260	0.1004	0.2971	0.2654	0.1025	6.5723	6.5201
Ego3DLM [†]	148.5	368.3	346.3	645.4	0.0190	0.9358	0.1036	0.3043	0.2706	0.1072	6.4837	5.917

Separate Motion/Text Decoding. Table D shows that jointly decoding motion and text in a single autoregressive sequence consistently improves accuracy and narration quality over separate motion and text decoding. Compared to the separated variant, joint decoding reduces motion prediction errors by 0.67% in APE, 0.51% in JPE, and 2.26% in FDE. It also improves future motion description, increasing Bleu-4 by 3.19% and Rouge_L by 2.42%, with retrieval gains of 1.96% in R@3 and 4.59% in R@1. Critically, motion-description alignment strengthens, with d_{pg} decreasing by 1.35% and d_{pp} dropping by 9.25%, supporting that joint generation yields more mutually grounded pose narration pairs.

Separate Past/Future Decoding. Table E shows that jointly decoding past and future tasks improves both forecasting and tracking compared to separate decoding. For motion prediction, joint decoding reduces JPE by 7.18%, with consistent improvements across the other forecasting metrics. For motion tracking, it yields larger gains, reducing APE by 28.04%, with consistent gains across the other tracking metrics. These results suggest that training the model to produce past and future motion within a unified generation process encourages shared temporal representations that benefit both tracking fidelity and forecasting performance. This joint supervision appears to strengthen the model’s understanding of motion dynamics, improving the fidelity of generated motion and enabling forecasting that more effectively leverages past observations to produce feasible future predictions.

Table E: Ablation study on joint vs. separate past and future task decoding. Joint decoding consistently improves both motion prediction and tracking.

Method	Motion Prediction (3 modes)						Motion Tracking				
	APE ↓	JPE ↓	ADE ↓	FDE ↓	FID ↓	Diversity ↑	APE ↓	Upper ↓	Lower ↓	J.A. ↓	Root ↓
Ego3DLM [†] (sep. past/future)	159.9	396.8	370.5	705.0	0.0335	0.9876	136.6	77.5	213.2	31.08	29.00
Ego3DLM [†]	148.5	368.3	346.3	645.4	0.0190	0.9358	98.3	53.4	155.3	22.5	19.62

Table F: Long-term motion prediction results across all methods. Ego3DLM[†] achieves state-of-the-art performance across all metrics, with the performance gap over baselines widening compared to the standard horizon.

Method	Motion Prediction (3 modes)					
	APE ↓	JPE ↓	ADE ↓	FDE ↓	FID	Diversity
Fiction	256.1	831.1	825.7	1,083.9	0.0896	0.3815
EgoLM (GT)	242.2	917.1	884.2	1,469.3	0.1086	0.2275
EgoLM (Inst. tune.)	238.6	922.3	887.4	1,472.9	0.1095	0.2231
UniEgoMotion	201.7	784.7	765.1	1,183.4	0.0858	0.3981
Ego3DLM [†]	172.2	638.5	618.3	1,079.8	0.0325	0.4141

Long-Term Motion Forecasting. Table F evaluates all methods on long-term motion prediction, where the increased temporal horizon amplifies the compounding effect of errors and scene-dependency of future motion. Ego3DLM[†] consistently outperforms all baselines across every metric, reducing JPE by 18.7% and ADE by 19.2% over the strongest baseline UniEgoMotion, demonstrating that our scene-aware training is particularly beneficial as the prediction horizon grows. Notably, EgoLM variants perform poorly even with ground-truth past motion as input, suggesting that neither language grounding nor GT conditioning alone is sufficient for accurate long-term forecasting without explicit 3D scene understanding.

3D Awareness and Physical Realism. We measure foot slide and foot contact following UniEgoMotion [13] to evaluate the 3D awareness and physical realism of the prediction in Table H. Since the Nymeria dataset does not provide explicit floor information, we select scenes with a flat floor and then measure foot slide and foot contact. We confirmed that our proposed Ego3DLM outperforms baselines across both metrics, demonstrating greater 3D awareness and physical realism.

Differences from the Original EgoLM.

The main reason for the difference from the metrics reported in the original EgoLM paper lies in the difference in the frequency and the tracking duration of

Table H: Evaluation of 3D awareness and physical realism using foot sliding and foot contact metrics.

Method	Foot Slide ↓	Foot Contact ↓
Fiction	26.01	0.0719
EgoLM	13.07	0.0517
UniEgoMotion	15.36	0.0480
Ego3DLM	8.91	0.0449

Table G: Comparison of our EgoLM reproduction with the original. Under the matched setting (60 Hz, 1 sec), our implementation closely matches the metrics reported in the original paper, while the gap under our main setting (10 Hz, 5 sec) is attributable to the difference in tracking frequency and duration.

Table 3	Setting	APE ↓	Upper ↓	Lower ↓	J.A. ↓	Root ↓
EgoLM (ours)	(10Hz, 5sec)	161.9	95.6	265.6	33.63	24.01
EgoLM (ours)	(60Hz, 1sec)	85.3	39.9	144.7	18.49	16.74
EgoLM (original)	(60Hz, 1sec)	73.3	49.6	124.5	12.48	13.23

motion data. While the original paper tracks for 1 second at 60Hz, we performed tracking for 5 seconds at 10Hz. Table G confirms that our implementation of EgoLM (original code unavailable) closely matches the metrics in the paper, demonstrating the fidelity of the reproduction and the validity of our Nymeria evaluation protocol. Because of differences in data composition, exact metric reproduction is infeasible, but our results closely match the reported values.

Multi-Hypothesis Motion Forecasting. Figure C visualizes the multi-hypothesis predictions generated by our model. Human motion is inherently non-deterministic, as a person may exhibit different plausible behaviors given the same past motion depending on their underlying intent. To address this property, we sample three distinct future predictions from the same model and examine the 3D human motion fidelity of each. Across all samples, the predicted trajectories remain plausible within the 3D scene, maneuvering through the indoor environment while avoiding or interacting with objects. These results indicate that our model effectively captures the non-deterministic nature of human motion intentions, enabled by its ability to leverage the rich semantic cues provided by the 3D scene information.

Table I: GRPO fine-tuning improves cross-modal fidelity. Rather than optimizing the alignment objective as a differentiable loss, applying it as a reward via GRPO further reduces both tracking and future matching distances.

Method (GPT-2, prediction)	Tracking Match. ↓	Future Match. ↓
Stage II w/ Motion-description alignment reward	5.6298	7.1120
Stage II	4.2861	5.9170
Stage II + Stage III (GRPO)	4.2571	5.8063

Usefulness of GRPO. The main goal of using GRPO is to holistically improve prediction/tracking performance along with their cross-modal fidelity. These cannot be achieved through next-token prediction with teacher forcing, as the ground-truth targets are fixed and already aligned by construction. Instead, the generation process itself must be optimized so that motion and text are accurately produced within alignment. As shown in Table I, GRPO finetuning yields stronger motion-description alignment than directly optimizing the same objective as a differentiable loss during instruction tuning. In addition, Table C

Table J: Cross-dataset generalization of models trained on Nymeria and evaluated on the Aria Digital Twin dataset.

Method	JPE ↓	ADE ↓	FDE ↓	APE(track.) ↓	Bleu-4 ↑
Fiction	586.5	551.7	870.2	268.7	–
EgoLM	460.8	420.5	727.0	237.9	0.0590
UniEgoMotion	467.0	481.0	734.8	338.2	–
Ours (Ego3DLM)	432.3	402.2	696.8	196.6	0.0628

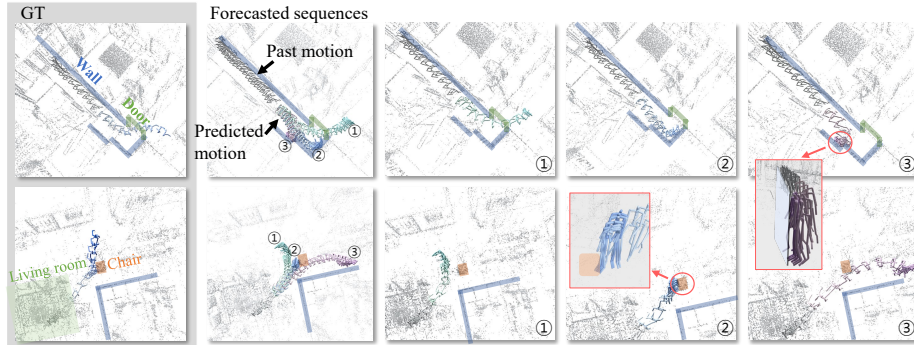


Fig. C: Examples of multi-hypothesis predictions. GT motion sequence (left), forecasted motion sequences (middle & right). All our predictions demonstrate feasible motion based on the 3D scene.

shows larger GRPO finetuning gains on a modern backbone, with a notable gain in matching distance compared to the GPT-2 backbone.

Cross-dataset Generalization. To verify the generalizability of the proposed Ego3DLM to other datasets, we evaluated models trained on Nymeria [10] on the Aria Digital Twin [11] in Table J. The degradation of zero-shot performance is minimal even under domain shift, showing our model’s generalizability. Our method again outperforms all baseline methods.

D Additional Qualitative Results

We present an additional qualitative comparison of our method against the ablation variant (Ours w/o 3D Scene) and baselines across diverse scenarios in Fig. D and Fig. E. The forecasted motion sequences from $t = 0s$ to $t = 5s$ are color-coded to indicate temporal progression, transitioning from white to blue, with the past motion shown in red. Ego3DLM demonstrates a robust understanding of environmental constraints. By explicitly conditioning on 3D scene features, our model predicts trajectories that capture the semantic intent while respecting spatial boundaries.

As shown in the Fig. E (bottom), the predicted motion paths accurately curve around obstacles and navigate through doorways, maintaining physically plausible

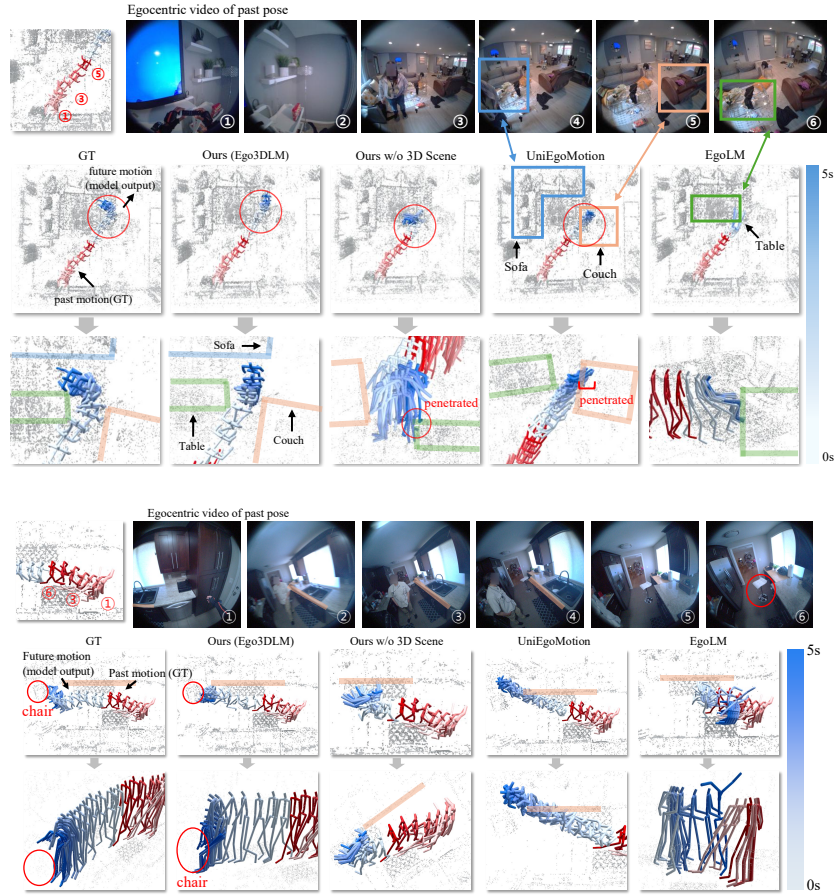


Fig. D: Qualitative results of motion prediction and future motion description. We visualize predicted motions from white to blue with temporal progression. Ego3DLM generates plausible trajectories that respect spatial boundaries, whereas baselines and the ablation variant frequently collide with scene geometry.

distances from surrounding structures. In contrast, our ablation variant, which lacks explicit 3D scene conditioning, generates motions that are kinematically coherent but environmentally inconsistent. EgoLM often produces trajectories that drift into obstacles, failing to align the agent’s orientation with the available free space. Similarly, while UniEgoMotion generates semantically plausible motions, it ignores the structural layout, leading to collisions. These results collectively highlight the superiority of Ego3DLM in forecasting future behaviors that are not only semantically aligned with the past egocentric video but also geometrically consistent with the complex 3D environment.

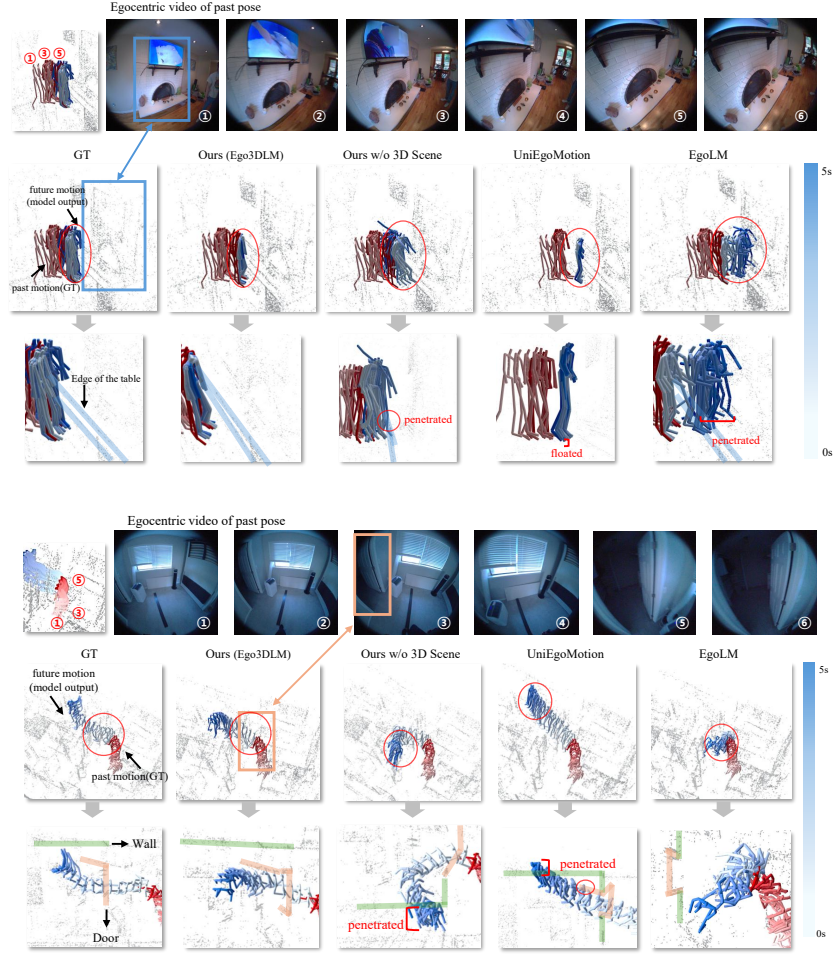


Fig. E: Qualitative results of motion prediction and future motion description. We visualize predicted motions from white to blue with temporal progression. Ego3DLM produces plausible trajectories that adhere to scene boundaries, while the baselines and ablated variant frequently intersect or collide with the surrounding geometry.

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